

Flattening Semantic Representations: Modelling Structural Ambiguity in Formal Semantics

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Abstract

Formal semantics traditionally assumes that the interpretation relation I is functional, leaving $\llbracket \cdot \rrbracket$ single-valued and treating structural ambiguity as an exception to be resolved by grammatical mechanisms. We argue this assumption is empirically inadequate: if grammar fully disambiguated linguistic structures, referential misunderstandings between interlocutors would be impossible. We propose to lift this presupposition by redefining I as a mathematical relation in the strict sense. The interpretation function $\llbracket \cdot \rrbracket^I$ is accordingly redefined as the function that, given a linguistic unit m and a relation I , collects the set E of all possible denotations of m under I — formally: $\lambda m. \lambda I. E$ s.t. $\forall x, \langle m, x \rangle \in I \Rightarrow x \in E$. We write $\llbracket m \rrbracket^I$ for this set and $\llbracket m \rrbracket^{I, \gamma}$ for the singleton selected by the choice function γ . For constituency ambiguities, the set $\llbracket S \rrbracket^I$ is generated via a flattening hypothesis: the semantic interpreter operates on linearized sequences $\text{LCA}(S)$ rather than on hierarchical structures directly. Since classical Functional Application (Heim and Kratzer, 1998) is undefined for sequences of $n > 2$ elements, a generalization FA^* is introduced, defined on n arguments with type-theoretic filtering built in. FA^* generates at most C_{n-1} bracketings (the Catalan numbers), yielding a set of denotations naturally. A choice function γ , argued to be primarily pragmatic in nature, selects a singleton from $\llbracket S \rrbracket^I$. Misunderstanding is modelled as the case where $\llbracket S \rrbracket_{\text{emitter}}^{I, \gamma} \neq \llbracket S \rrbracket_{\text{receiver}}^{I, \gamma}$. The full characterization of γ is left as an open research program.

Keywords: structural ambiguity, formal semantics, flat semantic representations, interpretation function, pragmatic disambiguation

1 Introduction

Ambiguity is an inherent and inevitable property of natural language (Cho, 2022; Dascal, 1999). Far from being a marginal defect, it pervades communication across domains: everyday conversation (Bazzanella and Damiano, 1999; Keysar and Henly, 2002), semantic interpretation (Siahaan et al., 2025), and even legal discourse (Schane, 2002). Crucially, Schane (2002) observes that legal systems have had to codify explicit procedures for resolving ambiguities in contracts and legislation — procedures grounded in pragmatic negotiation between the parties rather than in the application of grammatical rules. If language produced inherently unambiguous structures, such codification would be superfluous. This meta-institutional fact attests that ambiguity is a structural property of language, not an occasional processing failure.

Misunderstandings arising from structural ambiguity belong to the domain of performance and conversational interaction. Siahaan et al. (2025) show that ambiguity operates as a double-edged sword: while it enriches communicative expression, it also poses a systematic risk of misunderstanding when contextual cues are limited. Ambiguity resolution is therefore not — or not solely — a grammatical process.

Yet the tradition in formal semantics, from Heim and Kratzer (1998) onward, implicitly assumes that the interpretation relation I is functional, making $\llbracket \cdot \rrbracket^I$ single-valued. Ambiguity, when acknowledged, is treated as an exception to be resolved by dedicated grammatical mechanisms. This position is incompatible with the empirical facts above. The logical argument is simple: if $\llbracket S \rrbracket^I$ were always a singleton, then $\llbracket S \rrbracket_{\text{emitter}}^{I,\gamma} = \llbracket S \rrbracket_{\text{receiver}}^{I,\gamma}$ necessarily, making structural referential misunderstandings impossible — a prediction falsified by the evidence.

We propose to lift this presupposition. I is redefined as a mathematical relation, $\llbracket \cdot \rrbracket^I$ becomes set-valued, and a pragmatic choice function γ handles resolution. The present model focuses on *constituency ambiguities* — cases where a flat sequence admits multiple well-typed bracketings via FA*. Other types of structural ambiguity, involving distinct mechanisms such as covert movement or anaphora resolution, constitute a complementary research program that cannot be adequately addressed within the scope of this contribution.

The paper is organized as follows. Section 2 exposes the functionalist presupposition and its empirical limits. Section 3 presents the general theory of ambiguity, including the formal modeling of misunderstanding. Section 4 develops the flattening hypothesis and the FA* mechanism. Section 5 addresses

pragmatic resolution via γ . Section 6 concludes and opens perspectives.

2 The Functionalist Presupposition and Its Limits

2.1 The Tacit Presupposition in Formal Semantics

In the tradition initiated by Heim and Kratzer (1998), Functional Application (FA) is defined on binary syntactic tree nodes. For any branching node α with daughters β and γ' , $\llbracket \alpha \rrbracket = \llbracket \beta \rrbracket(\llbracket \gamma' \rrbracket)$ if $\llbracket \beta \rrbracket$ is a function whose domain contains $\llbracket \gamma' \rrbracket$ (and $\llbracket \gamma' \rrbracket(\llbracket \beta \rrbracket)$ otherwise). This definition presupposes a hierarchically structured input and produces exactly one output per structure, making I functional de facto.

The direct consequence is that $\llbracket m \rrbracket^I$ is necessarily a singleton for any well-formed expression m . Ambiguity, when recognized, is treated as an exceptional phenomenon to be resolved by specific mechanisms — scope constraints, syntactic disambiguation rules — rather than as a property to be modeled.

2.2 Empirical Limits

Constituency ambiguities are nonetheless widely attested and far from residual. The canonical example is (1):

John saw the man with the telescope. (1)

This sentence has two structurally distinct readings — PP attached to the DP or to the VP — both grammatically well-formed, with no grammatical mechanism capable of adjudicating between them.

If grammar fully disambiguated structures, referential misunderstandings of structural origin would be impossible, as both interlocutors would necessarily share the same semantic representation. Yet Keysar and Henly (2002) experimentally demonstrate that speakers systematically overestimate the transparency of their communicative intentions on syntactically ambiguous sentences: in their protocol, speakers estimated successful communication in 72% of cases, while actual receiver understanding stood at only 61%. Grammar does not guarantee convergence of interpretations.

Formalizing the contradiction: if $\llbracket S \rrbracket^I$ is a singleton for all S , then $\llbracket S \rrbracket_{\text{emitter}}^{I,\gamma} = \llbracket S \rrbracket_{\text{receiver}}^{I,\gamma}$ necessarily, making structural misunderstandings impossible — which contradicts the empirical record.

Complementary meta-institutional evidence comes from Schane (2002), who documents that legal systems have codified procedures for resolving

ambiguities in contracts and legislation. These procedures rest on pragmatic negotiation, not grammatical rule application, confirming that resolution belongs to performance rather than competence.

The functionalist presupposition is therefore empirically inadequate. The following section proposes a general theory that models ambiguity as a property of language and misunderstanding as a natural consequence.

3 A General Theory of Ambiguity

3.1 $\llbracket \cdot \rrbracket^I$ as a Set-Valued Function

We posit that I is a mathematical relation in the strict sense — a set of pairs $\langle m, x \rangle$ — and not a function. A single linguistic unit m may therefore be associated with multiple distinct denotations under I .

Accordingly, $\llbracket \cdot \rrbracket^I$ is redefined as:

$$\llbracket m \rrbracket^I = \lambda m. \lambda I. E \quad \text{s.t.} \quad \forall x, \langle m, x \rangle \in I \Rightarrow x \in E \quad (2)$$

$\llbracket m \rrbracket^I$ is thus the set E of all possible denotations of m under I , potentially containing multiple elements.

Two levels of interpretation are distinguished:

- $\llbracket \cdot \rrbracket^I \rightarrow \text{grammatical layer} \rightarrow E$ (set; models ambiguity)
- $\llbracket \cdot \rrbracket^{I, \gamma} = \gamma(\llbracket \cdot \rrbracket^I) \rightarrow \text{pragmatic layer} \rightarrow \text{singleton}$ (models resolution)

This dissociation is consistent with the observations of Sections 2: ambiguity resolution is not a grammatical process. Note that the Heim and Kratzer (1998) tradition corresponds to the degenerate case where $|\llbracket m \rrbracket^I| = 1$ for all m — i.e. where I is assumed to be functional. Our model is therefore a strict generalization of that framework, not a departure from it. It extends the formal architecture proposed in Akpoué (2025) by relaxing the implicit functionality assumption on I .

3.2 Formal Modeling of Misunderstanding

In the present model, emitter and receiver share the same grammatical representation $\llbracket S \rrbracket^I$ — ambiguity is in the grammar and is therefore shared by all interlocutors. The divergence lies in γ : each interlocutor applies their own choice function, sensitive to their individual cognitive and pragmatic context.

A misunderstanding arises when:

$$\llbracket S \rrbracket_{\text{emitter}}^{I,\gamma} \neq \llbracket S \rrbracket_{\text{receiver}}^{I,\gamma} \quad (3)$$

i.e. when the two interlocutors select distinct singletons from the same set $\llbracket S \rrbracket^I$. This formalization is directly compatible with Bazzanella and Damiano (1999), who define misunderstanding as a discrepancy between the meaning intended by the emitter and the meaning reconstructed by the receiver. It also accounts for Keysar and Henly (2002)’s finding that speakers overestimate the transparency of their intentions: speakers implicitly assume that γ_{receiver} will converge toward γ_{emitter} , which is not guaranteed.

4 Flat Semantic Representations and FA*

4.1 The Compositionality Problem

A naive approach to generating $\llbracket S \rrbracket^I$ would consist in directly assigning multiple interpretations to a given phrase, without going through its constituents. This would forfeit the benefits of decades of research in compositional semantics. A realistic model must compute the set of denotations through a mechanism grounded in the constituents, respecting the principle of compositionality.

If the syntactic hierarchy is preserved in the inputs of FA, however, only one result is obtained: Merge-based or X-bar systems determine a single order of FA application, yielding a single bracketing and a single denotation. This is precisely the functionalist presupposition we seek to lift. To obtain a set of denotations through a compositional mechanism, the hierarchy must be eliminated from semantic inputs.

4.2 The Flattening Hypothesis

We propose that a flattening layer operates between syntactic generation and the semantic interpreter, producing flat sequences from hierarchical structures. These sequences are inherently ambiguous: multiple bracketings can be reconstructed from them, yielding multiple denotations.

This hypothesis is further motivated by the observation that phonological interpretation also requires linearization: pronunciation operates on linear sequences, not on trees. A linearization algorithm therefore appears to operate at the output of syntactic concatenation, before phonological and semantic modules handle their respective processing — an elegant architectural unification of the two interfaces.

Two implementations of this flattening layer are conceivable. **Option A** posits an explicit intermediate function *LCA* between syntactic generation and the semantic interpreter. **Option B** integrates flattening directly into the semantic interpreter via a redefinition of $\llbracket \cdot \rrbracket^I$ such that $\llbracket \{A, \{B, C\}\} \rrbracket^I = \llbracket (A, B, C) \rrbracket^I$. We adopt **Option A**, which preserves modularity and avoids burdening the semantic interpreter with an operation that is not semantic in nature.

4.3 LCA as a Flattening Layer

LCA is the function that takes a hierarchical syntactic structure S as input and returns a flat sequence:

$$\text{LCA}(S) = (A, B, C, \dots) \quad (4)$$

$\llbracket \cdot \rrbracket^I$ is defined to operate always on $\text{LCA}(S)$, never on S directly. *LCA* is therefore a well-formedness condition on the input of $\llbracket \cdot \rrbracket^I$, not an optional implementation step.¹

4.4 FA^* as a Necessary Consequence

FA as defined in Heim and Kratzer (1998) is defined on binary syntactic tree nodes and is therefore undefined for flat sequences of $n > 2$ elements. A generalization is required. We introduce FA^* , defined on n arguments with type-theoretic filtering built in:

$$\llbracket \text{LCA}(S) \rrbracket^I = \text{FA}^*(A', B', C', \dots) \quad (5)$$

where $\text{FA}^*(A', B', \dots)$ is the set of all bracketings of the sequence $(A', B', \dots)$ for which *FA* is defined at every step (i.e. where all intermediate types are compatible). Type-theoretic filtering thus constitutes a first, grammatical layer of disambiguation: bracketings where *FA* is undefined due to type mismatch are automatically excluded. FA^* generates a set of denotations without additional stipulation.

The maximum number of bracketings generated by FA^* for a sequence of n elements is C_{n-1} , the $(n - 1)$ -th Catalan number:

¹The *TRANSFER*+*LCA* operation in the Minimalist Program (Chomsky, 1995; Collins and Stabler, 2016) naturally instantiates this role. A complete minimalist reformulation would, however, require revising the notion of denotation itself, which lies beyond the scope of this paper.

$$C_{n-1} = \frac{1}{n} \binom{2(n-1)}{n-1} \quad (6)$$

This gives the model a precise and testable prediction: ambiguity grows with the length of expressions, consistent with psycholinguistic data on the processing difficulty of long sentences. Table 1 illustrates this growth.

n (elements)	Max bracketings (C_{n-1})
2	1
3	2
4	5
5	14

Table 1: Maximum number of bracketings generated by FA* for sequences of n elements (Catalan numbers).

Returning to example (1), $\text{LCA}(S)$ yields the flat sequence (*saw, the-man, with-the-telescope*), with $n = 3$ and $C_2 = 2$ possible bracketings. FA* generates:

- $\text{FA}(\text{FA}(\text{saw}', \text{the-man}'), \text{with-the-telescope}')$: PP attached to the DP — *John saw [the man with the telescope]*
- $\text{FA}(\text{saw}', \text{FA}(\text{the-man}', \text{with-the-telescope}'))$: PP attached to the VP — *John [saw the man] with the telescope*

Both bracketings are type-theoretically well-formed; type-theoretic filtering does not disambiguate. Accordingly, $\llbracket S \rrbracket^I = \{d_1, d_2\}$, a two-element set, and γ is required for resolution.

5 Pragmatic Resolution of Ambiguity

5.1 Two Layers of Resolution

The type-theoretic filtering built into FA* constitutes a first, automatic disambiguation layer operating prior to γ . It reduces the set of possible denotations by excluding ill-typed bracketings. In cases where only one bracketing is well-typed, $\llbracket S \rrbracket^I$ is a singleton and γ is trivial — disambiguation is fully grammatical.

When $\llbracket S \rrbracket^I$ contains more than one element after filtering, γ operates on the residual ambiguity. This is the pragmatic layer: γ selects a singleton from $\llbracket S \rrbracket^I$ by drawing on extra-grammatical factors.

5.2 Properties and Factors of γ

γ is minimally defined as:

$$\gamma : \llbracket S \rrbracket^I \rightarrow \{d\} \quad \llbracket S \rrbracket^{I,\gamma} = \gamma(\llbracket S \rrbracket^I) \quad (7)$$

γ is extra-grammatical and specific to each interlocutor. It presumably aggregates several heterogeneous factors: the statistically most frequent interpretation in the speaker’s experience (Khan et al., 2012); structurally preferred or default readings (e.g. low PP attachment in English); priming effects from prior discourse; cognitive biases such as confirmation bias and speaker-perspective effects; and linguistic as well as extra-linguistic contextual cues.

The complete formal characterization of γ is intentionally left open at this stage of the research. γ aggregates factors belonging to pragmatics, psycholinguistics, and probabilistic computation; their precise interaction requires thorough empirical and interdisciplinary investigation before any formalization can be responsibly attempted.

6 Conclusion and Perspectives

We have proposed a model in which the interpretation relation I is a mathematical relation rather than a function, making $\llbracket \cdot \rrbracket^I$ set-valued. For constituency ambiguities, $\llbracket S \rrbracket^I$ is generated via the flattening hypothesis: the semantic interpreter operates on $\text{LCA}(S)$, and FA^* — a necessary generalization of heimkratzer1998 FA — produces the set of all well-typed bracketings, bounded by the Catalan numbers. Type-theoretic filtering provides a first, grammatical disambiguation layer; the choice function γ , primarily pragmatic in nature, handles residual ambiguity by selecting a singleton. Misunderstanding is modelled as the divergence $\llbracket S \rrbracket_{\text{emitter}}^{I,\gamma} \neq \llbracket S \rrbracket_{\text{receiver}}^{I,\gamma}$, a natural consequence of interlocutors applying distinct γ functions to a shared $\llbracket S \rrbracket^I$. This architecture cleanly separates what belongs to the grammar (generation of the denotation set) from what belongs to cognition and pragmatics (singleton selection).

Two perspectives are opened by this work. First, the formalization of γ constitutes an immediate interdisciplinary research program, drawing on

pragmatics, psycholinguistics (ambiguity resolution, processing time), and computational linguistics (word sense disambiguation, ambiguous parsing, packed representations). Second, if one adopts a componential semantics where lexemes have internal feature structure (Akpoué, 2025), the FA* mechanism can in principle be extended to the lexical level to model polysemy as low-level structural ambiguity; without such internal structure, this extension is inoperative and is left to future work.

References

- Kouamé Josué Akpoué. 2025. Introduction à la sémantique minimaliste. *Ziglôbitha, Revue des Arts, Linguistique, Littérature & Civilisations*, 3(16):319–338.
- Carla Bazzanella and Rossana Damiano. 1999. The interactional handling of misunderstanding in everyday conversations. *Journal of Pragmatics*, 31(6):817–836.
- Won Ik Cho. 2022. Ambiguity resolution in spoken language understanding. In *[TO COMPLETE: proceedings title]*. Reference to be completed and verified independently.
- Noam Chomsky. 1995. *The Minimalist Program*. MIT Press, Cambridge, MA.
- Chris Collins and Edward Stabler. 2016. A formalization of minimalist syntax. *Syntax*, 19(1):43–78.
- Marcelo Dascal. 1999. Introduction: Some questions about misunderstanding. *Journal of Pragmatics*, 31(6):753–762.
- Irene Heim and Angelika Kratzer. 1998. *Semantics in Generative Grammar*. Blackwell, Oxford.
- Boaz Keysar and Anne S. Henly. 2002. Speakers’ overestimation of their effectiveness. *Psychological Science*, 13(3):207–212.
- Imtiaz Khan, Roger P. G. van Gompel, Iain D. Gilchrist, and Martin J. Pickering. 2012. [to complete: exact title]. *Topics in Cognitive Science*. Reference to be completed and verified independently.

Sanford Schane. 2002. Ambiguity and misunderstanding in the law. *Thomas Jefferson Law Review*, 25(1):167–193. Volume and pages to be verified independently.

[FIRST NAME] Siahaan and 1 others. 2025. The role of semantic ambiguity in communication and misunderstanding. *[TO COMPLETE]*. Full reference to be completed and verified independently.